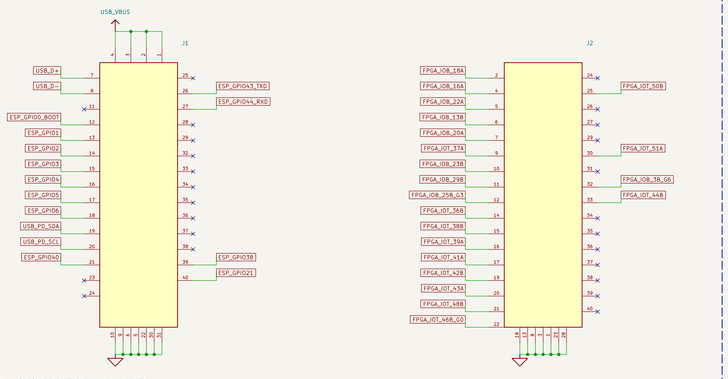
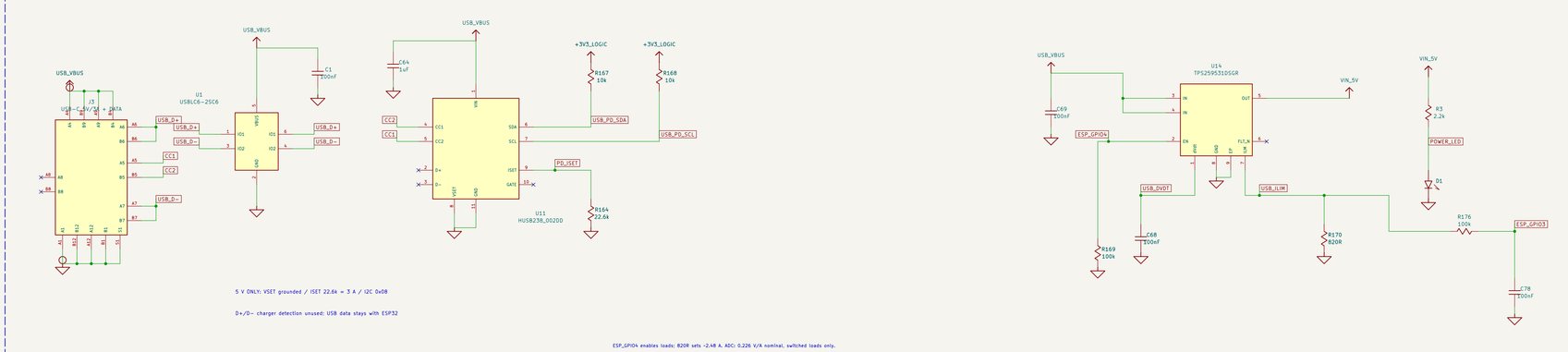
MODULE v1 / exact programmer mating interface



25 mm mating spacing / module top flush / no reset or boot switch

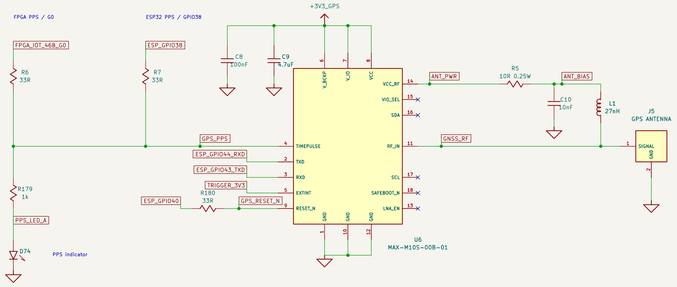
USB-C PD / 5 V, 3 A sink and protected power



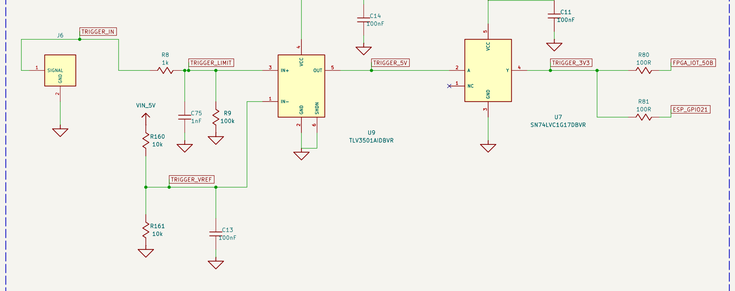
ESP_GPIO4 enables loads: R20R sets ~2.4A A; ADC: 0.226 V/A nominal, switched loads only.

Module and logic boot from raw 5 V. GPIO7/8 read PD status over I2C; firmware budgets current before enabling loads.

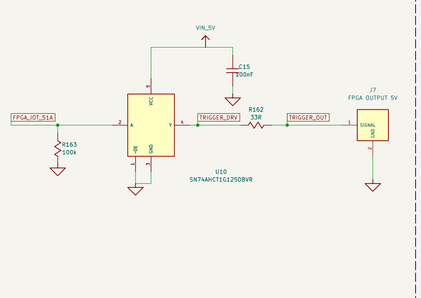
GNSS / PPS reference and active antenna bias



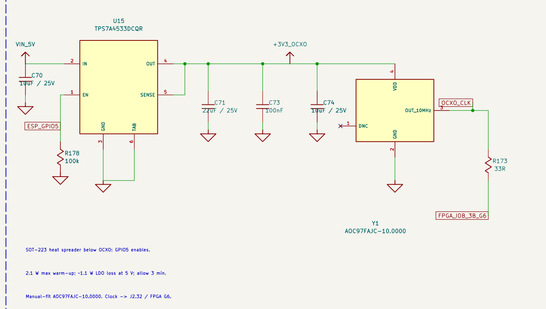
Trigger input / 5 V TTL / threshold ~2.5 V



Trigger output / 5 V TTL



10 MHz OCXO / 3.3 V linear supply

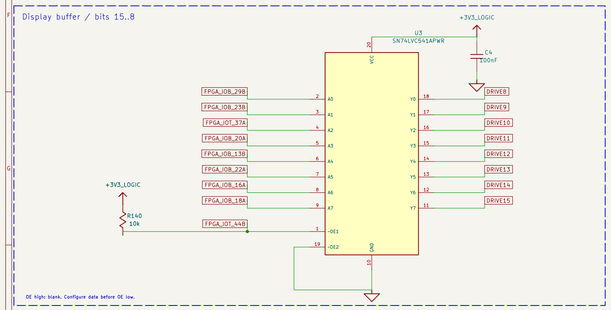


SOT-223 heat spreader below OCXO; GPIO5 enables.

2.1 W max warm-up; ~1.1 W LDO loss at 5 V: allow 3 min.

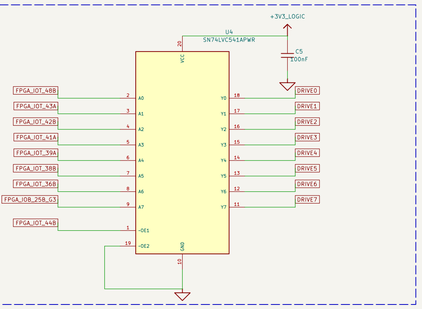
Manual-fit ASDC97FAJC-10.0000; Clock --> J2.32 / FPGA G6.

Display buffer / bits 15..8



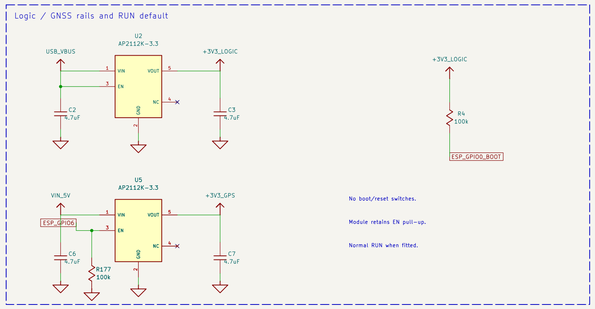
OE high: blank. Configure data before OE low.

Display buffer / bits 7..0



OE high: blank. Configure data before OE low.

Logic / GNSS rails and RUN default

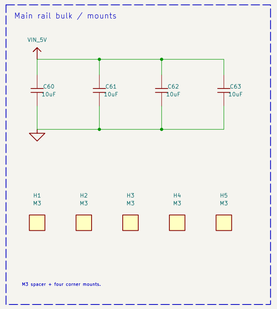


No boot/reset switches.

Module retains EN pull-up.

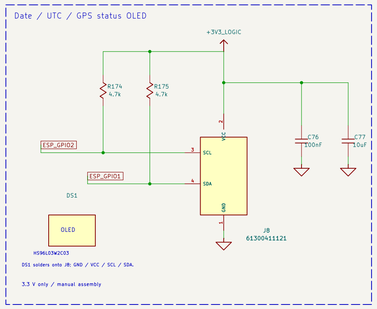
Normal RUN when fitted.

Main rail bulk / mounts



M3 spacer + four corner mounts.

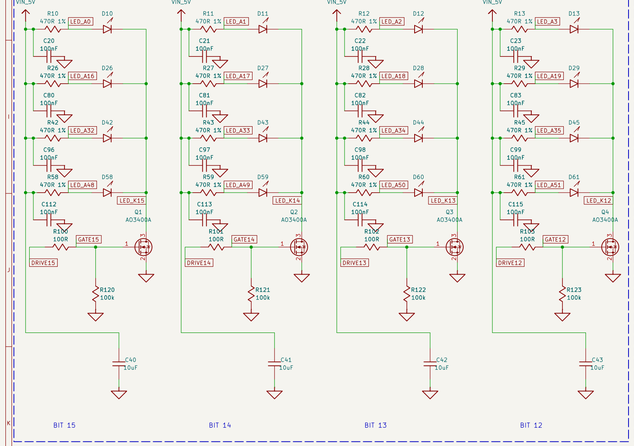
Date / UTC / GPS status OLED



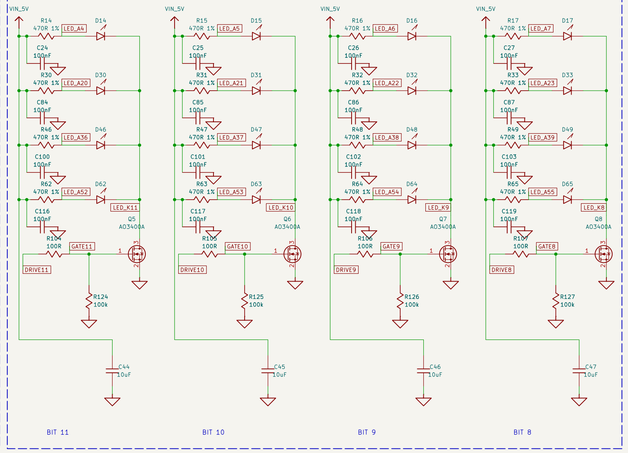
DS1 solders onto J8: GND / VCC / SCL / SDA.

3.3 V only / manual assembly

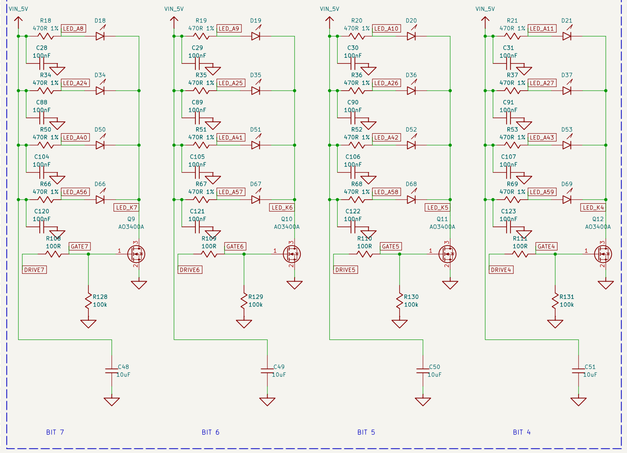
OPTICAL BANK 1 / bits 15..12 / four identical rows



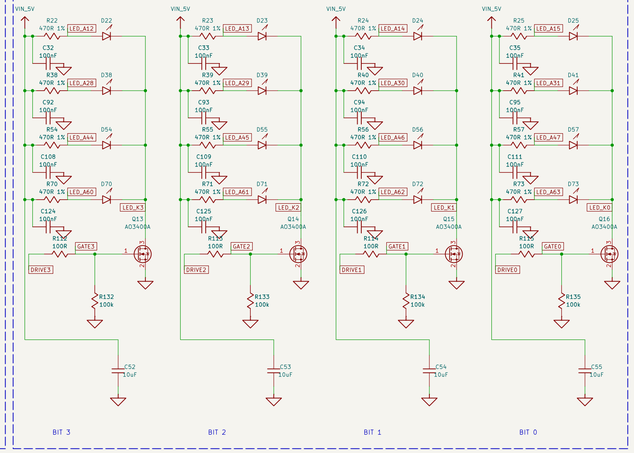
OPTICAL BANK 2 / bits 11..8 / four identical rows



OPTICAL BANK 3 / bits 7..4 / four identical rows



OPTICAL BANK 4 / bits 3..0 / four identical rows



Diagnostic probe pads / functional names on PCB / bare copper, no assembly parts

TP1 // VBUS (USB_VBUS)
TP2 // 5V_LOAD (VIN_5V)
TP3 // 5V_LOGIC (+5V_LOGIC)
TP5 // 3V3_GPS (+3V3_GPS)
TP6 // 3V3_OCXO (+3V3_OCXO)
TP7 // LOAD_EN (ESP_GPIO4)
TP8 // DCXO_EN (ESP_GPIO5)
TP9 // GPS_EN (ESP_GPIO6)
TP10 // PD_SDA (USB_PD_SDA)
TP11 // PD_SCL (USB_PD_SCL)
TP12 // LMON (ESP_GPIO3)
TP13 // PPS_SRC (GPS_PPS)
TP14 // PPS_FPGA (FPGA_IOT_46B_GO)
TP15 // PPS_ESP (ESP_GPIO38)
TP16 // CLK_FPGA (FPGA_IOB_3B_G6)
TP18 // TRIG_DVS (TRIGGER_OBEY)
TP19 // TRIG_CMP (TRIGGER_ICS)
TP20 // TRIG_3V3 (TRIGGER_3V3)
TP21 // OUT_FPGA (FPGA_IOT_51A)
TP22 // OUT_DRV (TRIGGER_DRV)
TP25 // GPS_TX (ESP_GPIO45_RXD)
TP26 // GPS_RX (ESP_GPIO43_TXD)
TP27 // GPS_RST_N (GPS_RESET_N)
TP30 // LED_OK_N (FPGA_IOT_49B)

bp-test
Sheet: /
File: calibrator.kicad_sch

Title: GPS optical time calibrator / module v1 carrier

Size: A1	Date:	Rev: A draft
KiCad E.D.A. 10.0.6		Id: 1/1